

Dated: _____ PROBABILITY AND STATISTICS NOTES

STATISTICS :- branch of mathematics, transform data into useful information for decision makers

→ **DESCRIPTIVE STATS** :- collecting, summarizing data using means, standard deviations etc

→ **INFERENCEAL STATS** :- drawing conclusions from data (population) by accounting for uncertainties and variation using a sample.

PARAMETER :- a numerical measure that describes a characteristic of a population

STATISTIC :- a numerical measure that describes a characteristic of a sample

DATA SOURCES :- Primary (data collector is data analyzer), secondary (data collector is not analyzer)
surveys, designed experiment, observational study, distributed by org or individual

DATA TYPES :- Categorical (Qualitative) represents groups (e.g. gender, colours, pass/fail)

Numerical (Quantitative) represents measurable values (e.g. height, weight, temperature)

→ Discrete : countable

→ Continuous : measured on a continuum

FORMULAE :- **GROUPED DATA**

Mean → $\mu = \frac{\sum fx}{\sum f}$ $\bar{x} = \frac{\sum fx}{\sum f}$
where fx corresponds to sum of freq $\times$ mid point of all intervals

Median → $L + \frac{H}{f} \left(\frac{n}{2} - CF \right)$
where L = lower class boundary
all of median class H = upper - lower class boundary
 f = frequency of median class
median class = $\frac{n}{2}$ CF = cumulative freq before median class

Mode class → class with highest frequency

Mode → $L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2} \right) \times H$

where f_m = freq of modal class

f_1 = freq of class before modal class

f_2 = freq of class after modal class

Variance → $\sigma^2 = \frac{\sum f(x - \mu)^2}{\sum f}$ $s^2 = \frac{\sum f(x - \bar{x})^2}{\sum f - 1}$
where x is mid point

Standard Deviation → $\sigma = \sqrt{\sigma^2}$ $s = \sqrt{s^2}$
C (width)
Flat / wide → high
tall / narrow → low

UNGROUPED DATA

Mean → $\mu = \frac{\sum x}{n}$ $\bar{x} = \frac{\sum x}{n}$

Median odd → middle element (sorted) $\frac{n}{2}$
even → $\left(\frac{n}{2} \right)^{th} + \left(\frac{n+1}{2} \right)^{th}$ element

Mode → most repetitions / highest freq

Variance → $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$ $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$

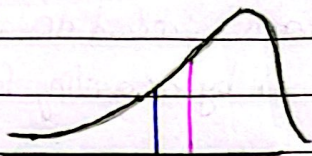
Standard Deviation → $\sigma = \sqrt{\sigma^2}$ $s = \sqrt{s^2}$

Coefficient of Variation → $CV = \frac{\sigma}{\mu} \times 100$
(percentage)

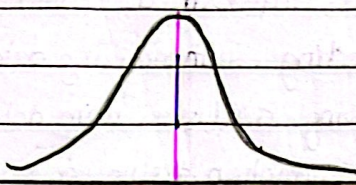
$CV = \frac{s}{\bar{x}} \times 100$

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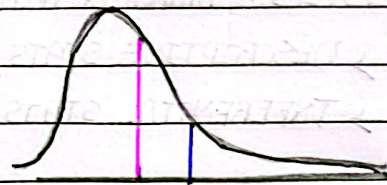
SHAPE OF A DISTRIBUTION



LEFT SKEWED
(mean < median)



SYMMETRIC
(mean = median)



RIGHT SKEWED
(mean > median)

QUARTILE MEASURES

First Quartile $Q_1 = \frac{n+1}{4}$ th

2nd Quartile $Q_2 = \frac{n+1}{2}$ th

3rd Quartile $Q_3 = \frac{3(n+1)}{4}$ th

$IQR = Q_3 - Q_1$

Upper Limit = $Q_3 + 1.5IQR$

Lower Limit = $Q_1 - 1.5IQR$

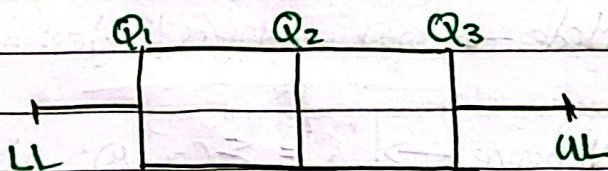
Whole number $\rightarrow$ use position directly

Fractional half $\rightarrow$ use avg of corresponding vals

Fractional num $\rightarrow$ e.g. = 5.25

= 5th + (6th - 5th) $\times$ 0.25

BOX PLOT



DATA REPRESENTATION (see slides)

CHAPTER 2 : PROBABILITY

Sample Space : set of all possible outcomes of a statistical experiment.

Sample Point : each individual outcome in the sample space

Tree Diagram : used for systematically listing sample points in sequential events

Rule Method :

① one task in m ways, another in n ways, cannot occur together $\rightarrow m+n$

② one task in m ways, another in n ways, total ways to do both $\rightarrow m \times n$

③ permutations (order matters)
• arranging n objects $\rightarrow n!$ ^{in order} then $\rightarrow (n-1)!$

• selecting and arranging r from $n \rightarrow \frac{n!}{(n-r)!}$

④ combinations (order doesn't matter) $\frac{n!}{r!(n-r)!}$
• selecting r objects from $n \rightarrow \frac{n!}{r!(n-r)!}$

⑤ Dividing n objects into r groups $\rightarrow \frac{n!}{n_1! n_2! n_3!}$

Dated: _____

Event: a subset of a sample space

Null Event: event with no elements

Complement: elements in S not in A

Mutually Exclusive: $A \cap B = \emptyset$

SET IDENTITIES

$$A \cap \emptyset = \emptyset$$

$$A \cup \emptyset = A$$

$$A \cap A^c = \emptyset$$

$$A \cup A^c = S$$

$$S^c = \emptyset$$

$$\emptyset^c = S$$

$$(A^c)^c = A$$

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Probability: the sum of the weights of all sample points in an event A . (the chance that an uncertain event will occur)

$$\begin{matrix} N - \text{total elements} \\ n - \text{corresponding to } A \end{matrix} \quad P(A) = \frac{n}{N}$$

$$0 \leq P(A) \leq 1, P(\emptyset) = 0, P(S) = 1$$

① additive rule:

• if A, B are mutually exclusive

$$P(A \cup B) = P(A) + P(B)$$

• if A, B are not mutually exclusive

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) -$$

$$P(A \cap B) - P(A \cap C) - P(B \cap C)$$

$$+ P(A \cap B \cap C)$$

② product rule:

• A and B are independent iff

$$P(A \cap B) = P(A)P(B)$$

→ if A, B IND: then not mutually exclus

• if A, B, C IND

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$P(A \cup B \cup C) = 1 - P(A' \cap B' \cap C')$$

• A, B independent iff

$$P(B|A) = P(B) \text{ OR } P(A|B) = P(A)$$

assuming the existence of conditional probabilities

• A, B dependent and can both occur

$$P(A \cap B) = P(A)P(B|A) \text{ provided } P(A) > 0$$

Conditional Probability: the probability of an event B occurring when it is known that some event A has occurred & prob B occurs given that A occurs)

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{provided } P(A) > 0$$

Baye's Rule: if the events B_i constitute a partition of S ($P(B_i) > 0$ and $P(A \cap B_i) \neq 0$)

$$P(B_r|A) = \frac{P(B_r \cap A)}{P(A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^n P(B_i \cap A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^n P(B_i)P(A|B_i)}$$

the event with the highest $P(B_r|A)$ is most likely cause of A for $r = 1, 2, \dots, n$

Dated: _____

CHAPTER 3: RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

RANDOM VARIABLE: a function that associates a real number with each element in the sample space

EXAMPLE: components defective / nondefective. Define the random var X by $X = \begin{cases} 0, & \text{if NO} \\ 1, & \text{if D} \end{cases}$

EXAMPLE: 4 red balls, 3 black balls, $\rightarrow$ taken out at a time from urn. The possible outcomes & the values of y of random var Y , where Y is the num of red balls are
 $X = \{RR(2), RB(1), BR(1), BB(0)\}$

DISCRETE SAMPLE SPACE: a set of distinct countable items

CONTINUOUS SAMPLE SPACE: a set of uncountable items, usually within a range.

DISCRETE PROBABILITY DISTRIBUTION: a discrete random var assumes each of its values with a certain prob.

EXAMPLE:

m	0	1	2
$P(M=m)$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{6}$

 } all possible outcomes } we can represent all probs of random var X through function $f(x) = P(X=x)$
 OR $f(x) = \frac{x^2+1}{x^2+0+1+2}$ } all up to 1

The set of ordered pairs $(x, f(x))$ is a probability function, probability mass function or probability distribution of the random var X , if for each possible outcome x :- $f(x) \geq 0$, $\sum_x f(x) = 1$, $P(X=x) = f(x)$

CONTINUOUS PROBABILITY DISTRIBUTION: a continuous random var has a prob of 0 for assuming exactly any

EXAMPLE: $f(x) = \begin{cases} 2x & -1 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$ of its values and hence associates probabilities with ranges.

The function $f(x)$ is a probability density function for the continuous random var X , defined over the set of real numbers if :- $f(x) \geq 0$ for all $x \in \mathbb{R}$, $\int_{-\infty}^{+\infty} f(x) dx = 1$, $P(a < X < b) = \int_a^b f(x) dx$

CUMULATIVE DISTRIBUTION FUNCTION

DISCRETE $\rightarrow F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$ for $-\infty < x < \infty$

CONTINUOUS $\rightarrow F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$ for $-\infty < x < \infty$

NOTE: to convert PDF back into PDF you differentiate

EXAMPLE PMF

x	0	1	2	3	4
$f(x)$	$\frac{1}{16} \cdot \frac{1}{4}$	$\frac{1}{16} \cdot \frac{1}{4}$	$\frac{1}{16} \cdot \frac{1}{4}$	$\frac{1}{16} \cdot \frac{1}{4}$	$\frac{1}{16} \cdot \frac{1}{4}$

$$F(0) = f(0) = \frac{1}{16}$$

$$F(1) = f(0) + f(1) = \frac{2}{16}$$

$$F(2) = f(0) + f(1) + f(2) = \frac{3}{16}$$

$$F(3) = f(0) + f(1) + f(2) + f(3) = \frac{4}{16}$$

$$F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1$$

$$F(x) = \begin{cases} 0, & \text{for } x < 0 \\ \frac{1}{16}, & \text{for } 0 \leq x < 1 \\ \frac{2}{16}, & \text{for } 1 \leq x < 2 \\ \frac{3}{16}, & \text{for } 2 \leq x < 3 \\ \frac{4}{16}, & \text{for } 3 \leq x < 4 \\ 1, & \text{for } x \geq 4 \end{cases}$$

EXAMPLE PDF

$$f(x) = \begin{cases} \frac{2x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$= \int_{-1}^x \frac{2t^2}{3} dt$$

$$= \frac{x^3 + 1}{9}$$

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{x^3 + 1}{9}, & -1 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

CONDITIONAL PROB IN CUMULATIVE DIST = $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Dated: _____

~~FOR JPD~~ FOR JPD $\rightarrow X, Y$ independent IF $f(x,y) = g(x)h(y)$

JOINT PROBABILITY DISTRIBUTIONS describes the probability of two (or more) random vars occurring together

JPMF $\rightarrow f(x,y) \geq 0$ for all (x,y)

$$\sum_x \sum_y f(x,y) = 1$$

$$P(X=x, Y=y) = f(x,y)$$

$$P[(X,Y) \in A] = \sum_{(x,y) \in A} f(x,y) \text{ for region } A$$

$f(x,y)$	0	1	2
0	$3/28$	$9/28$	$3/28$
1	$3/14$	$3/14$	0
2	$1/28$	0	0

$$g(x) = \sum_y f(x,y)$$

$$h(y) = \sum_x f(x,y)$$

for JPMF func's

x	0	1	2
$g(x)$	$16/28$	$15/28$	$3/28$

marginal distributions

y	0	1	2
$h(y)$	$15/28$	$6/14$	$1/28$

JPPF $\rightarrow f(x,y) \geq 0$ for all (x,y)

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$P[(X,Y) \in A] =$$

$$\iint_A f(x,y) dx dy$$

for any region A in the xy plane

$$f(x,y) = \begin{cases} \frac{2}{5}(2x+3y), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$\textcircled{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$\int_0^1 \int_0^1 \frac{2}{5}(2x+3y) dx dy = 1$$

$$= \frac{2}{5} + \frac{3}{5} = 1 \text{ shown}$$

$$\textcircled{3} \text{ given that } A = \{(x,y) | 0 < x < \frac{1}{2}, \frac{1}{4} < y < \frac{1}{2}\}$$

$$P[(X,Y) \in A] = \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} \frac{2}{5}(2x+3y) dx dy = \boxed{\frac{13}{160}}$$

to convert 2 vars into 1, we use conditional distrib

$$\text{conditional distribution of } Y \rightarrow f(y|x) = \frac{f(x,y)}{g(x)}, g(x) > 0$$

$$\text{conditional distribution of } X \rightarrow f(x|y) = \frac{f(x,y)}{h(y)}, h(y) > 0$$

same for continuous

EXAMPLE: conditional distribution of X given that Y=1,

$$f(0|1) = \frac{3/14}{3/7} = \frac{1}{2}$$

$$f(1|1) = \frac{3/14}{3/7} = \frac{1}{2}$$

$$f(2|1) = \frac{0}{3/7} = 0$$

x	0	1	2
$f(x 1)$	$\frac{1}{2}$	$\frac{1}{2}$	0

MARGINAL DISTRIBUTIONS

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 \frac{2}{5}(2x+3y) dy$$

$$= \frac{4x+3}{5}$$

$$h(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^1 \frac{2}{5}(2x+3y) dx$$

$$= \frac{2(1+3y)}{5}$$

CONDITIONAL DISTRIBUTION

$$f(y|x) = \frac{f(x,y)}{g(x)} \text{ for } 0 \leq x < y \leq 1$$

Dated: _____

CHAPTER 4: MATHEMATICAL EXPECTATION

MEAN / EXPECTED VALUE OF X RANDOM VAR

$$\begin{aligned} \rightarrow M &= E(X) = \sum x f(x) \text{ (discrete)} \\ \rightarrow M &= E(X) = \int_{-\infty}^{\infty} x f(x) dx \text{ (continuous)} \end{aligned}$$

EXPECTED VALUE OF FUNCTION $g(x)$ OVER X
RANDOM VAR WITH PROB DIST $f(x)$

$$\begin{aligned} \xrightarrow{\text{discrete}} M_g(x) &= E[g(x)] = \sum g(x) f(x) \\ \xrightarrow{\text{continuous}} M_g(x) &= E[g(x)] = \int_{-\infty}^{\infty} g(x) f(x) dx \end{aligned}$$

EXPECTED VAL OF JOINT PROB $f(x, y)$

$$M_{xy} = E[xy] = \sum \sum xy f(x, y) \text{ (discrete)}$$

EXPECTED VALUE OF FUNCTION $g(x, y)$ OVER X, Y

$$M_{g(x, y)} = E[g(x, y)] = \sum \sum g(x, y) f(x, y) \text{ (discrete)}$$

RANDOM VARS WITH JOINT PROB DIST $f(x, y)$

$$\xrightarrow{\text{continuous}} M_{g(x, y)} = E[g(x, y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

VARIANCE OF RANDOM VAR X WITH PROB
DIST $f(x)$ AND MEAN μ

$$\begin{aligned} \xrightarrow{\text{discrete}} \sigma^2 &= E[(X - \mu)^2] = \sum (x - \mu)^2 f(x) \\ \xrightarrow{\text{continuous}} \sigma^2 &= E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \end{aligned}$$

VARIANCE OF RANDOM VAR X $\xrightarrow{\text{ALTERNATIVE}^{\text{dist}} \text{cont}} \sigma^2 = E(X^2) - \mu^2$

FOR STANDARD DEVIATION $\rightarrow \sqrt{\sigma^2}$

VARIANCE OF FUNCTION $g(x)$ OVER X RANDOM
VAR WITH PROB DIST $f(x)$

$$\begin{aligned} \xrightarrow{\text{discrete}} \sigma_{g(x)}^2 &= E\{[g(x) - M_g(x)]^2\} \\ &= \sum [g(x) - M_g(x)]^2 f(x) \\ \xrightarrow{\text{continuous}} \sigma_{g(x)}^2 &= E\{[g(x) - M_g(x)]^2\} \\ &= \int_{-\infty}^{\infty} [g(x) - M_g(x)]^2 f(x) dx \end{aligned}$$

COVARIANCE OF X & Y RANDOM VARS WITH
JOINT PROB DIST $f(x, y)$

$$\begin{aligned} \xrightarrow{\text{discrete}} \sigma_{xy} &= E[(X - \mu_x)(Y - \mu_y)] \\ &= \sum \sum (x - \mu_x)(y - \mu_y) f(x, y) \\ \xrightarrow{\text{continuous}} \sigma_{xy} &= E[(X - \mu_x)(Y - \mu_y)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y) f(x, y) dx dy \end{aligned}$$

ALTERNATIVE

COVARIANCE OF X & Y RANDOM VARS WITH μ_x & μ_y $\rightarrow \sigma_{xy} = E(XY) - \mu_x \mu_y$

CORRELATION COEFFICIENT OF X & Y RANDOM VARS $\rightarrow \rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$

Dated: _____

CHAPTER 5: SOME DISCRETE PROBABILITY DISTRIBUTIONS

- BERNOULLI PROCESS:**
- ① the experiment consists of repeated trials
 - ② the outcome of each trial may be a success or a failure
 - ③ the probability of success (p) remains constant from trial to trial
 - ④ the repeated trials are independent

BINOMIAL DISTRIBUTION: $b(x; n, p) = {}^n C_x p^x q^{n-x}$

$p \rightarrow$ prob of success $n \rightarrow$ ind trials
 $q \rightarrow$ prob of failure $x \rightarrow x^{th}$ trial

- $p + q = 1$
- $\sum_{x=0}^n b(x; n, p) = 1$
- $P(a \leq x \leq b) = \sum_{x=a}^b b(x; n, p)$
- items are sampled with replacement + IND
- mean $\rightarrow \mu = np$
- variance $\rightarrow \sigma^2 = npq$
- standard deviation $\rightarrow \sigma = \sqrt{\sigma^2}$

HYPERGEOMETRIC DISTRIBUTION: $h(x; N, n, k) = \frac{{}^k C_x {}^{N-k} C_{n-x}}{{}^N C_n}$

$n \rightarrow$ sample size
 $N \rightarrow$ total items
 $k \rightarrow$ success items
 $N-k \rightarrow$ failure items
 $x \rightarrow$ no. of successes in sample size n

- $\max \{0, n - (N - k)\} \leq x \leq \min \{n, k\}$
- mean $\rightarrow \mu = \frac{n k}{N}$
- variance $\rightarrow \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} (1 - \frac{k}{N})$
- $\frac{k}{N} = p$ in binomial hence binomial may be viewed as a large-population version of hypergeometric.
- items are sampled without replacement + trials are not IND.

NEGATIVE BINOMIAL: $b^*(x; k, p) = {}^{x-1} C_{k-1} p^k q^{x-k}$

$p \rightarrow$ prob of success $k \rightarrow$ fixed no. of successes to achieve
 $q \rightarrow$ prob of failure
 $x \rightarrow$ trial no. on which k^{th} success occurs

- $x = k, k+1, k+2, \dots$
- trials repeated until a fixed number of successes occur
- independent trials
- if $k=1$, then turns into geometric distribution

GEOMETRIC DISTRIBUTION: $g(x; p) = p q^{x-1}$

$p \rightarrow$ prob of success $x \rightarrow$ trial on which first success occurs
 $q \rightarrow$ prob of failure

- $x = 1, 2, 3, \dots$
- mean $\rightarrow \mu = \frac{1}{p}$
- variance $\rightarrow \sigma^2 = \frac{1-p}{p^2}$
- independent trials

POISSON DISTRIBUTION: $p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$

$\lambda \rightarrow$ avg num of outcomes per unit time
 $t \rightarrow$ time interval / specified region
 $x \rightarrow$ no. of events occurring in t

- $x = 0, 1, 2, \dots$
- $P(x, t) = \sum_{x=0}^{\infty} p(x; \lambda t)$

Dated: _____

- mean & variance $\rightarrow \mu = \sigma^2 = \lambda t$
- events in different time intervals / regions do not affect each other (no memory)
- chance of event happening in short time / small space depends only on length / size
- In short time / small space, it's almost impossible for multiple events to happen at once

APPROXIMATION OF BINOMIAL DISTRIBUTION BY A POISSON DISTRIBUTION:

$$b(x; n, p) \xrightarrow{n \rightarrow \infty} p(x; \mu) \quad \text{Binomial to Poisson Dist Approx}$$

- $n \rightarrow \infty$ (no. of trials very large) ($n > 50$)
- $p \rightarrow 0$ (prob of success very small) ($np < 5$)
- $np \xrightarrow{n \rightarrow \infty} \mu$ remains constant

Dated: _____

CHAPTER 6: SOME CONTINUOUS PROBABILITY DISTRIBUTIONS

NORMAL DISTRIBUTION:
$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$\mu \rightarrow$ mean
 $\sigma \rightarrow$ standard deviation
 $x \rightarrow$ value of whose prob density is to be calculated

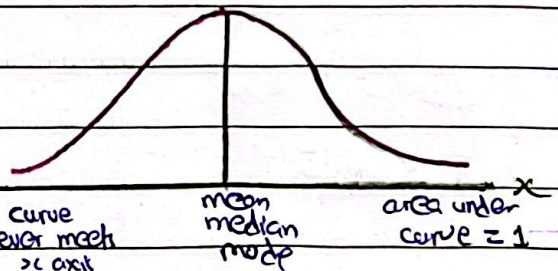
$-\infty < x < \infty$

• bell-shaped, symmetric, unimodal

• mean = median = mode

• $Z = \frac{X - \mu}{\sigma}$ z score is the no.

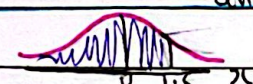
of standard deviations value x is from μ



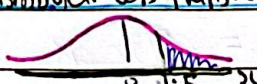
• if $\mu = 0$ and $\sigma^2 = 1$ then standard normal distribution

• all normally distributed vars transformed into standard normally distributed vars using Z .

CASES $\rightarrow$

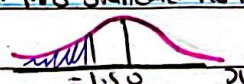


$P(Z < 1.5) = 0.9332$



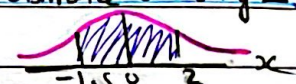
$P(Z > 1.5) =$

$1 - P(Z < 1.5)$



$P(Z < -1.5) =$

$P(Z > 1.5)$



$P(Z < 2) - P(Z < -1.5)$

$P(-1 < Z < 1) = 2P(Z < 1) - 1$

NORMAL APPROXIMATION TO BINOMIAL:

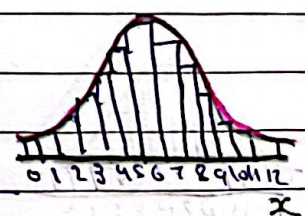
NOT INCLUDED

• $np \geq 5$ & $nq \geq 5$

• $+0.5$ = continuity correction

• $\approx$ area under normal curve left of $x+0.5$

$$P(X \leq x) = \sum_{k=0}^x b(k; n, p) = P\left(Z \leq \frac{x+0.5 - np}{\sqrt{npq}}\right)$$



$P(X \geq 12) = P(X > 11) = P(X \geq 11.5)$

$P(X \leq 12) = P(X < 13) = P(X \leq 12.5)$

$P(X > 12) = P(X \geq 13) = P(X > 12.5)$

$P(X < 12) = P(X \leq 11) = P(X < 11.5)$

CENTRAL LIMIT THEOREM: if many random samples are taken from a population, the avg of those samples will start to look like a normal dist, even if original data isn't normal

• if $n \geq 30$, non-normal to normal distribution conversion.

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

STANDARD ERROR OF MEAN:

$$SE = \frac{\sigma}{\sqrt{n}}$$

(for samples drawn with replacement OR without replacement from infinite population)

tells how much sample mean fluctuates from true population mean

$$SE = \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}$$

(if large sample taken from small population without replacement)

Dated: _____

CHAPTER 9 - ESTIMATION

STATISTICAL INFERENCE: Involves drawing conclusions about population parameters based on sample data

- **ESTIMATION**: determining likely values for parameters info only from sample data $\left\{ \begin{array}{l} \text{CLASSICAL} \\ \text{BAYESIAN} \end{array} \right\}$ prior knowledge + sample data
- **HYPOTHESIS TESTING**: making decisions about presumed characteristics

CLASSICAL ESTIMATION METHODS:

① **POINT ESTIMATE**: a single value used to estimate an unknown population parameter, a statistic is unbiased if the expected value of estimator = true population parameter (for e.g. sample variance; sample SD is biased)

② **INTERVAL ESTIMATION**: gives a range likely to contain the parameter (smallest)
a $100(1-\alpha)\%$ CI has $(1-\alpha)$ probability of containing true value where α = significance level.

$$P(\hat{\theta}_{LL} < \theta < \hat{\theta}_{UL}) = 1 - \alpha \quad (\text{confidence coefficient})$$

TRADE OFF: wider CI less precision, narrower CI low confidence

SINGLE SAMPLE:

ESTIMATING MEAN WHEN σ^2 KNOWN $\rightarrow$ $\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 $n \geq 30$

ONE-SIDED CONFIDENCE BOUNDS ON μ , σ^2 KNOWN $\rightarrow$ $\bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}}$ (upper)
 $\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}$ (lower)

ASSUMPTIONS: ① the sample is a random sample

② either $n \geq 30$ or the population is normally distributed if $n < 30$

ERROR: ERROR IN ESTIMATION $\rightarrow |\bar{X} - \mu| \leq z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ "how far off" limit

SAMPLE SIZE FOR ERROR MARGIN e $\rightarrow n = \left(\frac{z_{\alpha/2} \cdot \sigma}{e} \right)^2$ sample size needed for e error.

ESTIMATING MEAN WHEN σ^2 UNKNOWN $\rightarrow \bar{X} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}}$
 $n < 30$ $\nu = n - 1$

TWO SAMPLES

ESTIMATING DIFFERENCE B/W 2 MEANS WHEN σ_1^2 & σ_2^2 KNOWN $\rightarrow (\bar{X}_1 - \bar{X}_2) \pm z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
 $(\mu_1 - \mu_2)$

ESTIMATING DIFFERENCE B/W 2 MEANS WHEN σ_1^2 & σ_2^2 UNKNOWN BUT EQUAL $\rightarrow (\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
 $(\mu_1 - \mu_2)$

Dated: _____

$$V = n_1 + n_2 - 2, \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

pooled sample variance

ESTIMATING DIFFERENCE B/W 2 MEANS WHEN σ_1^2 & σ_2^2 UNKNOWN AND NOT EQUAL $\rightarrow$

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$V = \left[\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right]^2 \frac{(n_1 - 1)(n_2 - 1)}{[(S_1^2/n_1)^2(n_1 - 1) + (S_2^2/n_2)^2(n_2 - 1)]}$$

PAIRED OBSERVATIONS DIFFERENCE CONFIDENCE INTERVAL FOR $\mu_D = \mu_1 - \mu_2 \rightarrow$

$$\bar{d} - t_{\alpha/2} \frac{S_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2} \frac{S_d}{\sqrt{n}} \quad V = n - 1$$

PREDICTION INTERVALS: estimate the range where a future single observation may fall, with the mean

PREDICTION INTERVAL WHEN σ^2 KNOWN $\rightarrow \bar{X} - z_{\alpha/2} \sigma \sqrt{1 + \frac{1}{n}} < x_0 < \bar{X} + z_{\alpha/2} \sigma \sqrt{1 + \frac{1}{n}}$

PREDICTION INTERVAL WHEN σ^2 UNKNOWN $\rightarrow \bar{X} - t_{\alpha/2} S \sqrt{1 + \frac{1}{n}} < x_0 < \bar{X} + t_{\alpha/2} S \sqrt{1 + \frac{1}{n}} \quad (V = n - 1)$

TOLERANCE INTERVALS: estimate an interval that contains a specific proportion of the population (e.g. 95%) with a given confidence interval (e.g. 95%)

proportion $\rightarrow (1 - \alpha)$

confidence level $\rightarrow (1 - \gamma)$

$$\bar{X} \pm k \cdot s$$

$s \rightarrow$ sample s.d

$\bar{X} \rightarrow$ sample mean

$k \rightarrow$ factor from table

Dated: _____

CHAPTER 10: HYPOTHESIS TESTING

STATISTICAL HYPOTHESIS: an assertion or conjecture concerning one or more populations

SINGLE MEAN FORMULAE

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} ; \sigma \text{ KNOWN } (n \geq 30)$$

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} ; v = n - 1 ; \sigma \text{ UNKNOWN}$$

CRITICAL VAL APPROACH

- establish AR and CR
- Find if z/t calc lies in AR or CR

P VALUE APPROACH

- find z/t calc
- find prob ($z > \dots$) ^{or t}
- compare with α prob

TWO MEAN FORMULAE

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} ; \mu_1 - \mu_2 = d_0 ; \sigma_1 \neq \sigma_2 \text{ KNOWN}$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}} ; v = n_1 + n_2 - 2 \quad n < 30$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

; $\sigma_1 = \sigma_2$ AND UNKNOWN

$$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} ; v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

; $\sigma_1 \neq \sigma_2$ AND UNKNOWN

PAIRED SAMPLE T TEST

$$t = \frac{\bar{d} - d_0}{s_d / \sqrt{n}} ; v = n - 1$$

where $\bar{d}$ sample ^{mean} difference
 d_0 hypothesized mean diff
 M_d population mean diff

Dated: _____

CHAPTER 11: SIMPLE LINEAR REGRESSION AND CORRELATION

REGRESSION: dependence of one variable (dependent) on another (independent)

$$\hat{y} = \beta_0 + \beta_1 x + \varepsilon$$

$\hat{y}$ - response β_0, β_1 - reg coeffs (intercept & slope)
 x - regressor ε - random error w $E(\varepsilon) = 0$
constant σ^2

deterministic $\rightarrow y$ exactly determined by x ($\varepsilon = 0$)

probabilistic $\rightarrow y$ varies even for fixed x

This model assumes the expected value of y , given x , lies on straight line (true reg).

random errors make it deviate

to find estimated regression line $\rightarrow \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

$$\hat{\beta}_0 = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$\hat{\beta}_1 = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

SIR provides an estimate of the population regression line

CORRELATION: describes the strength and direction of a relationship between two variables

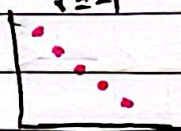
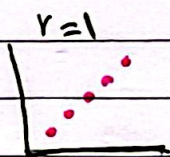
lies between $\rightarrow -1 \leq r \leq 1 \rightarrow r=0$ means no linear relationship

• symmetric $r_{xy} = r_{yx}$

• unit free

• r for sample ρ for population

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$



when $r \neq 0$, either there really is a relationship OR it just happened by chance in this sample, to confirm use hypothesis testing for ρ

$$t = r \sqrt{\frac{n-2}{1-r^2}} \quad \nu = n-2$$

• if H_0 rejected, either

$\rightarrow X$ directly causes Y

$\rightarrow Y$ directly causes X

$\rightarrow$ OR 2 third var causes both

$\rho = 0$ linearly uncorrelated

$\rho > 0$ pos lin correlated

$\rho < 0$ neg lin correlated

$\rho \neq 0$ linearly correlated

COEFFICIENT OF DETERMINATION $\rightarrow R^2 = (r)^2$

• measures ~~explains~~ how well a regression model explains variability in outcome

• varies from 0 (no predictive power) to 1 (perfect prediction)

ANOTHER WAY OF FINDING REGRESSION COEFFICIENTS: GRADIENT DESCENT

Dated: _____

EXAMPLE QUESTION

x	y	$\alpha = 0.05$
2	3	number of iterations = 3
3	6	compute loss function at each iteration
4	7	initial coefficients (0,0)

STEPS

$$MSE = \text{Loss Function} = \frac{1}{2n} \sum (b_0 + b_1x - y)^2$$

$$MSE = \frac{1}{2(3)} [C b_0 + 2b_1 - 3)^2 + C b_0 + 3b_1 - 6)^2 + C b_0 + 4b_1 - 7)^2]$$
$$= \frac{1}{6} [C b_0 + 2b_1 - 3)^2 + C b_0 + 3b_1 - 6)^2 + C b_0 + 4b_1 - 7)^2]$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{n} \sum (C b_0 + b_1x - y)$$

$$\frac{\partial L}{\partial b_0} = \frac{1}{3} [C b_0 + 2b_1 - 3) + C b_0 + 3b_1 - 6) + C b_0 + 4b_1 - 7)]$$
$$= \frac{1}{3} (3b_0 + 9b_1 - 16)$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{n} \sum (C b_0 + b_1x - y)x$$

$$\frac{\partial L}{\partial b_1} = \frac{1}{3} [C b_0 + 2b_1 - 3)2 + 3(C b_0 + 3b_1 - 6) + 4(C b_0 + 4b_1 - 7)]$$
$$= \frac{1}{3} (9b_0 + 29b_1 - 52)$$

$$b_0^{new} = b_0^{old} - \alpha \frac{\partial L}{\partial b_0}$$
$$b_1^{new} = b_1^{old} - \alpha \frac{\partial L}{\partial b_1}$$

$$b_0^{new} = b_0^{old} - \frac{0.05}{3} (3b_0^{old} + 9b_1^{old} - 16)$$

Dated: _____

$$b_1^{\text{new}} = b_1^{\text{old}} - \frac{0.05}{3} (9b_0^{\text{old}} + 29b_1^{\text{old}} - 52)$$

STARTING ITERATIONS WITH $(0, 0)$

$$\textcircled{1} b_0 = 0 - \frac{0.05}{3} (3(0) + 9(0) - 16) = 0.26667$$

$$b_1 = 0 - \frac{0.05}{3} (9(0) + 29(0) - 52) = 0.86667$$

$$L = \frac{1}{6} [(0.26667 + 2(0.86667) - 3)^2 + (0.26667 + 3(0.86667) - 6)^2 + (0.26667 + 4(0.86667) - 7)^2]$$

$$= 3.58145$$

$$\textcircled{2} b_0 = 0.26667 - \frac{0.05}{3} (3(0.26667) + 9(0.86667) - 16) = 0.39$$

$$b_1 = 0.86667 - \frac{0.05}{3} (9(0.26667) + 29(0.86667) - 52) = 1.27445$$

$$L = \frac{1}{6} [(0.39 + 2(1.27445) - 3)^2 + (0.39 + 3(1.27445) - 6)^2 + (0.39 + 4(1.27445) - 7)^2]$$

$$= 0.91377$$

$$\textcircled{3} b_0 = 0.39 - \frac{0.05}{3} (3(0.39) + 9(1.27445) - 16) = 0.446$$

$$b_1 = 1.27445 - \frac{0.05}{3} (9(0.39) + 29(1.27445) - 52) = 1.46663$$

$$L = \frac{1}{6} [(0.446 + 2(1.46663) - 3)^2 + (0.446 + 3(1.46663) - 6)^2 + (0.446 + 4(1.46663) - 7)^2]$$

$$= 0.32474$$

INFERRING SLR COEFFICIENTS

CONFIDENCE INTERVALS $\rightarrow \boxed{\beta_0: b_0 - t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2} < \beta_0 < b_0 + t_{\alpha/2} \frac{s}{\sqrt{n S_{xx}}} \sqrt{\sum_{i=1}^n x_i^2}}$

$\boxed{v = n - 2}$

$\boxed{S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n}}$

$\boxed{\beta_1: b_1 - t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s}{\sqrt{S_{xx}}}}$

T-STATISTIC $\rightarrow \boxed{\beta_0: T = \frac{b_0 - \beta_0}{s \sqrt{\sum_{i=1}^n x_i^2 / (n S_{xx})}}}$

$\boxed{\beta_1: T = \frac{(b_1 - \beta_1) / (s / \sqrt{S_{xx}})}{s / s} = \frac{b_1 - \beta_1}{s / \sqrt{S_{xx}}}}$

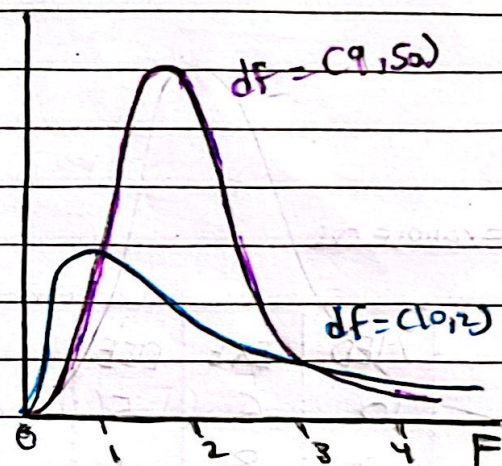
$= \frac{b_1 - \beta_1}{s / \sqrt{S_{xx}}}$

Dated: _____

CHAPTER 16 - ANALYSIS OF VARIANCE

F DISTRIBUTION

- right skewed curve (F curve)
- two degrees of freedom: ① numerator df
② denominator df
- hence since there are infinitely many df pairs, there are infinitely many F-distributions with different shapes
- total area under F curve = 1
- starts at 0 on horizontal axis and extends indefinitely to the right, never touching horizontal axis.



ANOVA → used to compare means of 3 or more groups (population means)

ASSUMPTIONS:

- ① simple random samples
- ② independent samples
- ③ normal populations
- ④ equal population standard deviations

STEPS

- ① $H_0: \mu_1 = \mu_2 = \dots = \mu_k$
 H_1 : Not all means are equal (at least 1 different)
- ② decide α (significance level)
- ③ compute values SSA, SSE, F-test

source of variation	sum of squares	degrees of freedom	mean square	computed F
treatments	SSA	$k-1$	$S_1^2 = \frac{SSA}{k-1}$	$\frac{S_1^2}{S_2^2}$
errors	SSE	$1(n-1)$	$S_2^2 = \frac{SSE}{1(n-1)}$	
total	SST	$1(n-1)$		

- ④ find F table using dfs and α
- ⑤ compare & accept / reject hypothesis.

Dated: _____

EXAMPLE Q3

	MED	EXE	DTE
	10	6	5
	12	8	9
	9	3	12
	15	0	8
	13	2	4
$\bar{X}$	$\frac{59}{5} = 11.8$	$\frac{19}{5} = 3.8$	$\frac{38}{5} = 7.6$
S_i^2	5.7	10.2	10.3

$$\bar{GM} = \frac{11.8 + 3.8 + 7.6}{3} = 7.7333$$

$k = 3$ (populations)

n = num of observations in group

$N = 15$ (total observations)

$$\alpha = 0.05$$

$$\begin{aligned} \text{d.f. } N &= k - 1 = 2 \\ \text{d.f. } D &= N - k = 12 \end{aligned}$$

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : atleast one different

$$SSA = \text{b/w group variance} = \boxed{S_B^2 = \frac{\sum n_i (\bar{x}_i - \bar{x}_{GM})^2}{k - 1}}$$

$$SSA = \frac{5(11.8 - 7.7333)^2 + 5(3.8 - 7.7333)^2 + 5(7.6 - 7.7333)^2}{3 - 1} = 80.07$$

$$SSE = \text{within group variance} = \boxed{S_W^2 = \frac{\sum (n_i - 1) S_i^2}{\sum (n_i - 1)}}$$

$$SSE = \frac{4(5.7) + 4(10.2) + 4(10.3)}{4 + 4 + 4} = 8.73$$

$$F\text{-test} = \boxed{F = \frac{S_B^2}{S_W^2}}$$

$$F = \frac{80.07}{8.73} = 9.17$$

$$F_{\text{TABLE}} = 3.89$$

$$9.17 > 3.89$$

hence H_0 rejected

